

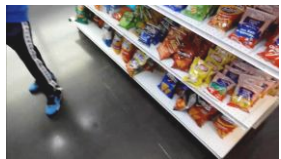
Video Stream

t_0

t_{k-2}

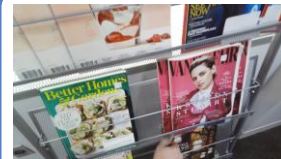
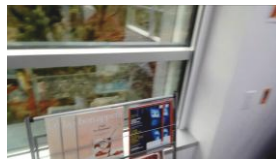
t_{k-1}

t_k



Query: What magazine is the man taking?

...



...

Sampled video frames

Sliding window video segment

Instruction-Guided Proposal Parser

Proposals:

- A hand holding a magazine with colorful cover
- A man with a magazine in hand walk away from a magazine rack
- ...

Lightweight Proposal Matching Module

Video Embeds

...



Text-video Similarity score



Text Embed

Off-the-shelf MLLM

Respond & Check

Assistant: The man is taking Living Magazine.